

Reflection on My Data Analysis Journey

My Journey

Over four weeks and eight tasks, this internship took me through the full data analysis lifecycle. I started with the fundamentals, exploring a sales dataset in Pandas and learning to clean messy data by handling missing values, removing duplicate rows, and fixing incorrect data types. These early tasks taught me that trustworthy analysis depends on disciplined data preparation before any real insight is possible.

In the following weeks I moved into professional tools and deeper techniques. I built my first interactive Power BI dashboard, analyzed trends through time-series work with rolling averages and anomaly detection, and strengthened my SQL skills with joins, aggregations, subqueries, and window functions. I also learned statistical rigor through A/B testing, stating hypotheses clearly and judging significance with proper tests rather than trusting surface-level differences.

The capstone brought everything together: SQL extraction, a Chi-Square hypothesis test, an interactive dashboard, and a stakeholder presentation. My biggest lesson was that technical results matter only when they are communicated clearly and honestly, including their limitations.

Two Advanced Skills I Want to Learn Next

Looking ahead, I want to deepen two advanced areas. First, **machine learning**, moving into ensemble methods, hyperparameter tuning, and model interpretability. Second, **deep learning**, understanding neural networks and their applications to image and language data.